1

Logger

LOGGER

You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error". Implement the

```java
interface Logger {
    void debug(String message);
    void info(String message);
    void warning(String message);
    void error(String message);
}

public class Log4jLogger implements Logger {

    public void debug(String message) {
        System.out.println("[LOG4J DEBUG] " + message);
    }

    public void info(String message) {
        System.out.println("[LOG4J INFO] " + message);
    }

    public void warning(String message) {
        System.out.println("[LOG4J WARNING] " + message);
    }
```

Your INPUT goes here!

OUTPUT

```
[LOG4J DEBUG] Debugging application.
[LOG4J INFO] Application started successfully.
[LOG4J WARNING] Low disk space.
[LOG4J ERROR] Database connection failed.
```

2.

Character

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement the interface in classes

```java
interface Character {
    void move();
    void attack();
    void takeDamage();
}

public class Hero implements Character {

    public void move() {
        System.out.println("Hero moves.");
    }

    public void attack() {
        System.out.println("Hero attacks.");
    }

    public void takeDamage() {
        System.out.println("Hero takes damage.");
    }
```

Your INPUT goes here!

OUTPUT

```
Hero moves.
Hero attacks.
Hero takes damage.
```

3.

Sort

SORT

Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.

```java
import java.util.*;
public class main{
  public static void main(String[]args){
    Scanner sc=new Scanner (System.in);
    ArrayList<Integer>list=new ArrayList<Integer>();
    System.out.println("enter 5 integers:");
    for(int i=0;i<5;i++){
      list.add(sc.nextInt());
    }
    Collections.sort(list);
    System.out.println("sorted list:"+list);
  }
}
```

50
20
40
10
30

OUTPUT

```
enter 5 integers:
sorted list:[10, 20, 30, 40, 50]
```

4.

QUESTIONS 8 / 8 completed

HashSet

HASHSET

Write a Java program to create a HashSet of strings and remove a specific element from it.

PROGRAM EDITOR

Java

Run Save

```java
import java.util.*;

public class Main {
    public static void main(String[] args) {

        HashSet<String> set = new HashSet<>();

        set.add("Apple");
        set.add("Banana");
        set.add("Mango");
        set.add("Orange");
        set.add("Grapes");

        System.out.println("Before Removal: " + set);

        set.remove("Mango");

        System.out.println("After Removal: " + set);
    }
}
```

INPUT

Apple
Banana
Mango
Orange
Grapes
Mango

OUTPUT

5.

TreeMap

TREEMAP

Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

```java
import java.util.*;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        TreeMap<String, Integer> map = new TreeMap<>();

        System.out.println("Enter 3 names and marks:");

        for (int i = 0; i < 3; i++) {
            String name = sc.next();
            int marks = sc.nextInt();
            map.put(name, marks);
        }

        System.out.println("TreeMap Entries:");

        for (Map.Entry<String, Integer> entry : map.entrySet()) {
            System.out.println(entry.getKey() + " : " + entry.getValue());
        }
```

OUTPUT

```
Enter 3 names and marks:
TreeMap Entries:
arun : 85
bala : 95
```

```
ram 90
arun 85
bala 95
```

6.

PriorityQueue

PRIORITYQUEUE

Write a Java program to create a PriorityQueue of integers and add elements to it.

```java
import java.util.*;
public class main{
  public static void main(String[]args){
    Scanner sc = new Scanner(System.in);
    PriorityQueue<Integer>pq=new PriorityQueue<Integer>();
    System.out.println("enter 5 integers:");
    for(int i=0;i<5;i++){
      pq.offer(sc.nextInt());
    }
    System.out.println("Priority Queue :"+pq);
  }
}
```

OUTPUT

```
enter 5 integers:
Priority Queue :[10, 20, 40, 50, 30]
```

```
50
20
40
10
30
```

7.

LinkedHashMap

LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

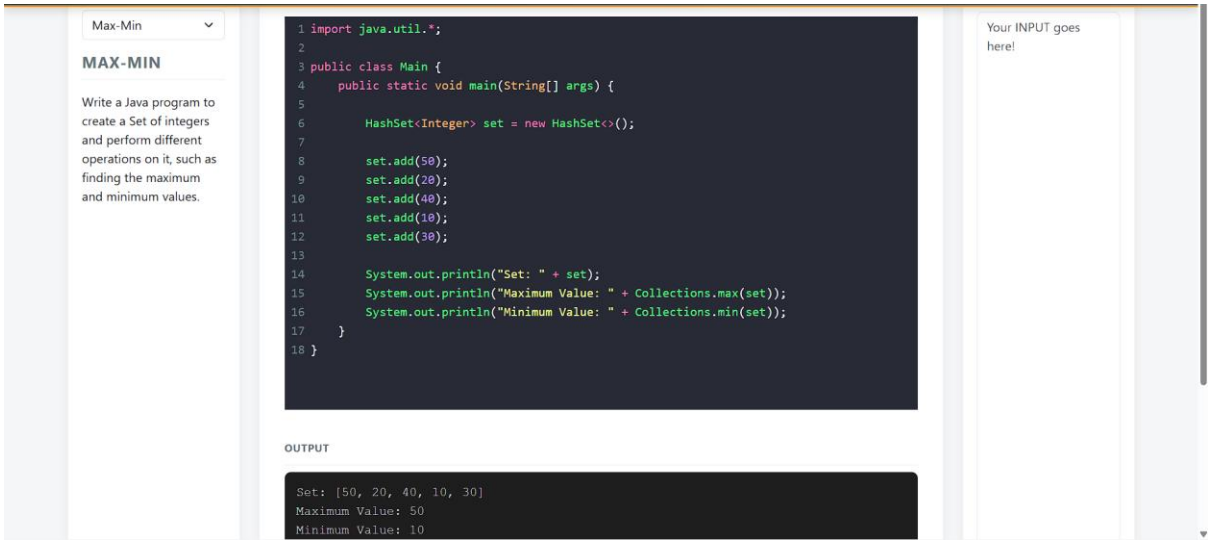
```java
import java.util.*;

public class Main {
    public static void main(String[] args) {

        LinkedHashMap<String, Integer> map = new LinkedHashMap<>();

        map.put("Ram", 90);
        map.put("Arun", 85);
        map.put("Bala", 95);

        System.out.println("Keys in insertion order:");

        for (String key : map.keySet()) {
            System.out.println(key);
        }
    }
}
```

OUTPUT

```
Keys in insertion order:
Ram
Arun
Bala
```

```
Ram 90
Arun 85
Bala 95
```

8.


Max-Min
MAX-MIN
Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.
1 import java.util.*;
2
3 public class Main {
4 public static void main(String[] args) {
5
6 HashSet<Integer> set = new HashSet<>();
7
8 set.add(50);
9 set.add(20);
10 set.add(40);
11 set.add(10);
12 set.add(30);
13
14 System.out.println("Set: " + set);
15 System.out.println("Maximum Value: " + Collections.max(set));
16 System.out.println("Minimum Value: " + Collections.min(set));
17 }
18 }
OUTPUT
Set: [50, 20, 40, 10, 30]
Maximum Value: 50
Minimum Value: 10
Your INPUT goes here!